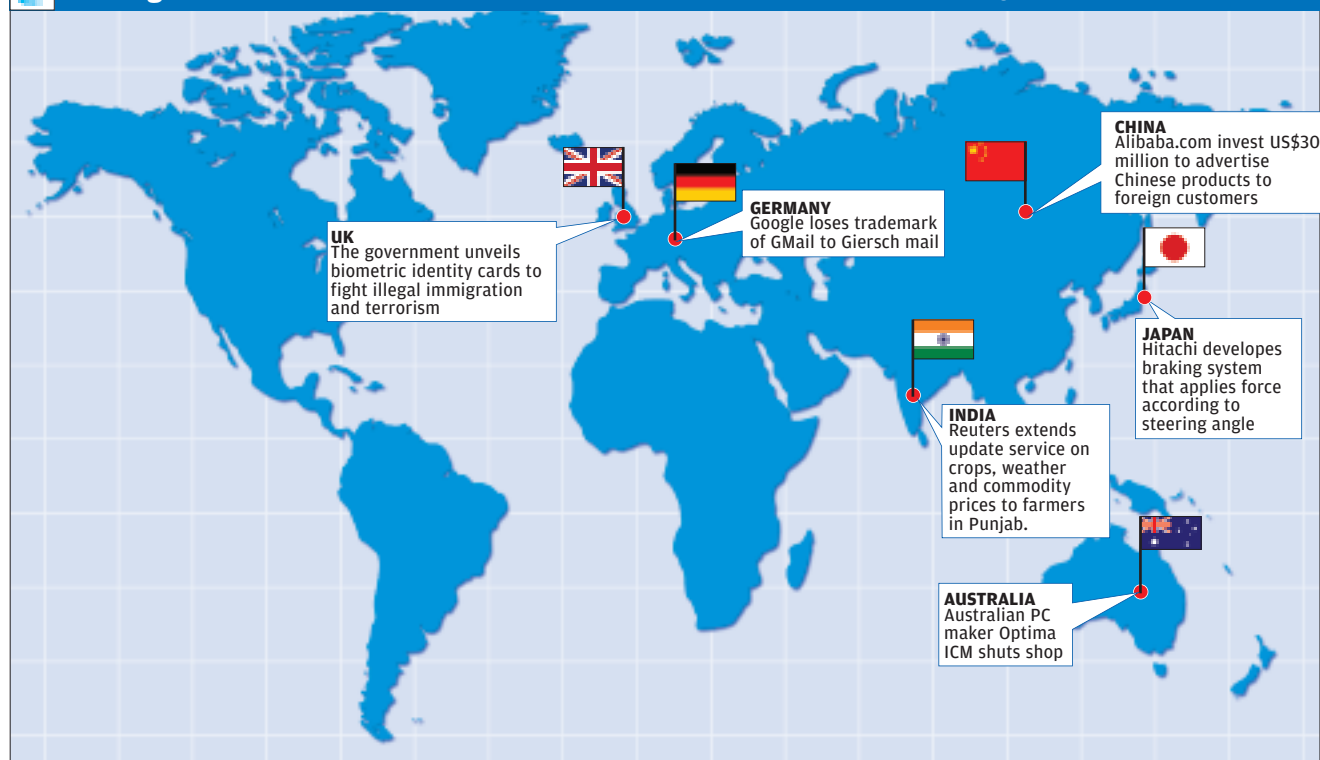




TISSUE-JET TECHNOLOGY?

Printing Hearts

Decades ago, space missions seemed to be a dream, or fiction. Then one fine day, President Kennedy felt man should reach the moon. The objective was realised, and soon fiction was reality! That's precisely how inventions change our perspective of ideas.

When we think about printers, all that comes to mind are offices, pictures, documents, and most of all paper. But one Japanese professor went the Kennedy way and thought out of the box. Although the idea seems impossible when you think about it, he's actually inching closer to achieving it.

Professor Makoto Nakamura is working on a printer that prints organs.

No, not the musical instruments, or pictures of body parts, but actual organs like hearts and kidneys! He has already succeeded in printing a tube of cells as narrow as the human hair. Nakamura says it will take him about 20 years before he is able to 'print' a heart. These tubes were made by an inkjet printer that Nakamura's

team has developed after three years of work.

The printer is precise to the extent of one-thousandth of a millimetre. A tube of three millimetres takes a minute and a half to print using Nakamura's printer.

The journey began 13 years ago, when Professor Nakamura was a paediatrician. During those

days, he came across several children with threatening heart problems that didn't respond to conventional medication. It deeply saddened Nakamura to see them die without any options. A 36-year-old Nakamura felt he shouldn't just be a mute spectator and should do something about it. He left a decade long practice to become a researcher and contribute to progress in this field.

One day Nakamura was researching artificial hearts and realised that mechanical hearts are not effective substitutes to donor organs, because mechanical organs are not able to generate energy, secrete hormones or even fight infection. In his search, he discovered that an inkjet printer's drops and human cells are about the same size — approximately one-hundredth of a millimetre.

Nakamura got in touch with a customer care

